Puzzle - Exploring the subway

The main puzzle of the game is to find where the goal platform is in relation to the player. This is mainly done through trial-and-error, but there are some patterns in the generation that can aid the player. For example, every transfer station is from an X line to an O line, or vice versa. Never X to X, or O to O. Another consistency is the length of the line corresponding to the color (green lines have 5 stations, blue lines have 6, red lines have 7, and black lines have 8. The final helpful tidbit is that X lines have transfers at every even number station that isn't the end of the line, and same for O lines but for off numbers. My goal for making this puzzle was to have rules for generation, so as to eventually add an map generator. With a random map, you could then find the target platform more easily with a little bit of quick thinking and a good mental map.

The biggest inspiration for this puzzle was a book series called *Dungeon Crawler Carl*, specifically book 3, *The Dungeon Anarchist's Cookbook*. The main premise of that book is a massive train puzzle, where the station numbering and train naming is one of the central mysteries that need solving. I couldn't make anything similar to the named lines in the book, nor is there as much difficulty surviving on the trains.

Originally, I had trouble thinking what I wanted the cues for what each line did to be. I thought about tying in shapes, and have that be the number of stations indicator, but that seemed too on-the-nose, so I scrapped it. Color is something I knew I wanted to do something with. My first thought was that lines of the same color couldn't cross, but I didn't have enough colors in the game to make that possible, so it got folded into the X and O system. Color ended up being the number of stations instead of the shapes, to make it a bit more obfuscated. Finally, I really wanted to tie in something from *Dungeon Crawler Carl*, and that ended up being the transfer stations. Far from a one-to-one copy, but the idea that the platform number tells you a little about what is at each station is preserved.

My sketch of the map (not comprehensive, I didn't have a blue marker):

